

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2025

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. Let S_n be the group of permutations of n objects (call them $1, 2, \dots, n$). A group G is called a *transitive permutation group of degree n* if G is isomorphic to a subgroup Γ of S_n , and whenever $1 \leq i, j \leq n$, there exists $g \in \Gamma$ such that $g(i) = j$.
 - (a) (2 points) Suppose that G is a transitive permutation group of degree n . Regard G as a subgroup of S_n . For $1 \leq i \leq n$, let G_i be the set of elements of G fixing i . Show that for each i, j , the groups G_i and G_j are conjugate (that is, there exists $g \in G$ such that $G_i = gG_jg^{-1}$).
 - (b) (1 point) Show that $G_1 \cap G_2 \cap \dots \cap G_n = \{1\}$.
 - (c) (3 points) Suppose $K \subseteq G_1$ and K is normal in G . Show that $K = \{1\}$.
 - (d) (4 points) Let A be a group containing a subgroup H of index n . Suppose that $K = \{1\}$ is the only subgroup $K \subseteq H$ such that K is normal in A . Show that A is a transitive permutation group of degree n .
2.
 - (a) Let M be an $n \times n$ complex matrix of rank r , with $0 \leq r \leq n$. Show that X^{n-r} is a factor of the characteristic polynomial of M .
 - (b) Let W be the $n \times n$ matrix all of whose entries are 1's. Find the characteristic polynomial of W .
3. Let R be a PID and let M be a finitely generated R -module. Let F be a field containing R . Show that $\text{Hom}_R(M, F)$ (that is, R -module homomorphisms from M to F) and $M \otimes_R F$ have the same dimension as vector spaces over F .

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4. Let R be a commutative ring with 1. The ring R is said to be Artinian if every descending chain of ideals stops. That is, for each sequence

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

of ideals of R , there exists n such that $I_n = I_{n+1} = \cdots$.

- (a) Show that if R is Artinian and I is an ideal of R , then R/I is an Artinian ring.
 - (b) Show that if R is an Artinian integral domain, then R is a field. (*Hint:* Consider the ideals $(x) \supseteq (x^2) \supseteq (x^3) \cdots$.)
 - (c) Show that if R is Artinian and P is a prime ideal of R , then P is a maximal ideal.
 - (d) Show that if $P_1, \dots, P_n$ are distinct maximal ideals of a commutative ring R (with 1), then there exists an element $x \in P_1 \cap \cdots \cap P_{n-1}$ with $x \notin P_n$.
 - (e) Show that an Artinian ring has only finitely many prime ideals.
5. Let p be a prime and let L/K be a finite Galois extension of fields of degree p . Let σ generate $\text{Gal}(L/K)$. Assume that there exists a primitive p th root of unity ζ in K . Let $\alpha \in L$ and assume

$$\delta = \alpha + \zeta^{-1}\sigma(\alpha) + \zeta^{-2}\sigma^2(\alpha) + \cdots + \zeta^1\sigma^{p-1}(\alpha) \neq 0.$$

- (a) Show that $\delta \notin K$.
 - (b) Show $\delta^p \in K$.
 - (c) Show $L = K(\delta)$.
6. (a) Let S_3 be the permutation group on 3 objects. List the conjugacy classes in S_3 and compute the character table of S_3 .
- (b) Define a representation ρ of S_3 by letting S_3 act on $\mathbb{C}^3$ by permutation of coordinates:

$$\rho(\sigma)(x_1, x_2, x_3) = (x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}).$$

Determine the character of ρ and find its decomposition into a sum of irreducible characters found in part (a).